
Correct Scores, Wrong Scientific Conclusions: Auditing AI Forecasts of Chaotic Dynamics

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Abstract

1 Learned forecasters increasingly stand in for scientific models that are themselves
2 imperfect, and a forecast score alone cannot say whether the surrogate succeeds
3 for the right reasons. We test five conditions that connect a score to a scientific
4 claim: held-out generalization, complete failure accounting, matched intervention,
5 structural identifiability, and stability across systems. We apply these tests to
6 k -pendulum ($k \in \{1, 2, 3\}$) and Lorenz-63 forecasts from language models, a
7 time-series foundation model, learned dynamics, sparse identification, and clas-
8 sical integrators. Each test changes a substantive conclusion. Learned-model
9 error rises from about 0.26 rad on training trajectories to 0.97–1.14 rad on held-
10 out trajectories, where linearized dynamics and nearest-neighbor lookup perform
11 better. Retaining failed forecasts lowers the reported performance of models with
12 incomplete coverage. An unmatched comparison says that hiding physical con-
13 stants helps; a paired intervention instead finds a +0.069 rad penalty. Two exact
14 scale symmetries prove that the absolute constants are not identifiable from angle
15 trajectories. Increasing the Neural ODE training set improves error from 1.090
16 to 0.861 rad but does not reach known-equation integration. The pendulum or-
17 dering also fails to transfer to Lorenz-63, where SINDy recovers the polynomial
18 dynamics to numerical precision. These results show that a correct leaderboard
19 can still support incorrect scientific conclusions. We release the prompts, outputs,
20 checkpoints, seeds, and compute records needed to repeat the audit.

21 1 Introduction

22 Scientific claims require more than low forecast error. When a learned forecaster stands in for an
23 imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched
24 inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of
25 the surrogate. We study these failure modes directly. Given state $\mathbf{s}_t$, the task is to predict $\mathbf{s}_{t+T}$.
26 The k -link pendulum is integrable at $k=1$ and chaotic for $k \geq 2$, has high-precision ground truth,
27 and permits controlled changes to physical metadata. Lorenz-63 provides a second system with a
28 different observation model and hypothesis class.

29 Our contribution is a claim audit for dynamical-system forecasts. It tests held-out generalization,
30 counts every failed forecast, records the information available to each method, changes disclosure
31 on identical trajectories, checks identifiability before scoring parameter recovery, varies the training
32 budget, and tests whether conclusions transfer to a second system. Table 1 states the conclusion
33 before and after each test. The central result is empirical and direct: all five tests alter or reject a
34 conclusion supported by the unaudited comparison. Three of the five are the familiar concerns of
35 working with imperfect models, applied to the evaluation itself: misspecification (a Hamiltonian prior
36 imposed on damped dynamics), identifiability (exact parameter degeneracies in the observation map),
37 and transfer (a hypothesis class that contains one system and not the other).

Table 1: Conclusions before and after the claim audit. Each row changes one part of the evaluation while preserving the relevant model outputs or comparison set.

Test	Unaudited conclusion	Audited conclusion
Generalization	Learned dynamics forecast accurately.	Error rises about fourfold; simple baselines perform better.
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Disclosure	Withholding constants improves forecasts.	Withholding constants increases pooled error by 0.069 rad.
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler-Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models such as Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024] are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We evaluate these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Here the output is a numerical state at a specified time in a chaotic trajectory.

Data-driven identification and abstention. SINDy fits sparse governing equations from measured trajectories [Brunton et al., 2016]; nonlinear vector autoregression provides a compact reservoir-computing formulation [Gauthier et al., 2021]. Structural identifiability asks which parameter combinations can be recovered from observed outputs, including under parameter symmetries [Borgqvist et al., 2024]. Selective prediction ordinarily reports risk together with coverage [Geifman and El-Yaniv, 2017]. Here failed cells remain in the main score because models are compared at fixed task coverage.

3 Benchmark Design

Choice of systems. We use the k -link pendulum to vary the number of coupled links while retaining a high-precision numerical solution. Its constants can also be hidden without changing the underlying trajectory, which permits the matched test in §5.2. The smaller Lorenz-63 experiment repeats the local-model comparison on a second chaotic system. We did not rerun the LLM or disclosure experiments on Lorenz.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the

Table 2: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned; learned hidden cells withhold hidden constants.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except <code>hidden</code>	—

73 RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is a numerical reference
 74 by construction, not an independent competitor. Euler and the symplectic integrator are distinct
 75 known-physics methods. Inputs available to each family are stated in §3.3.

76 3.2 Regimes and grid

77 Three regimes test generalization to non-standard constants (Table 8). In `changed_hidden` the
 78 constants (g, L, m , damping) are not disclosed to the model. This is a matched metadata-disclosure
 79 test; a single state contains no observed dynamics from which to identify the constants. An evaluation
 80 cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s,
 81 modalities $\{\text{coords, image, image+coords}\}$ (LLMs only), and prompting $\{\text{direct, chain-of-thought}\}$
 82 (LLMs only). Non-LLM models use the coordinate/direct setting.

83 **Out-of-sample protocol.** Learned dynamics models are evaluated off their training data. We
 84 generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions
 85 (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed
 86 by `movement_id`. The cross-family comparison (§5.1) uses horizons $\{1, 10\}$ s, where chaos has
 87 begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model. We report no
 88 in-sample numbers and quantify the size of the in-sample effect directly (§5.1).

89 3.3 Inputs and grouping

90 Table 2 states what each family receives. Chronos is a time-series model, so we evaluate it in its
 91 standard form: a history-conditioned forecaster with a 10 s pre- $t=0$ context window. The LLMs and
 92 learned dynamics models are state-conditioned from $t=0$. For learned models, hidden-regime cells
 93 are equalized by withholding hidden constants and using normal-regime prior constants. A separate
 94 asymmetry runs in the opposite direction: the LLMs are large-scale proprietary models, whereas
 95 the learned models are small (256-wide, 3-layer MLPs trained on a single 8 GB GPU; Appendix B).
 96 The behavior of larger physics-informed networks is outside the scope of this study. We separate
 97 known-physics methods (RK4, Euler, symplectic) from black-box methods (LLMs, Chronos, learned
 98 dynamics); the former are numerical references.

99 3.4 Metrics

100 The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over
 101 links and trajectories. We also track Cartesian bob error, angular-velocity sign match, and energy
 102 drift.

103 **Reliability-adjusted error.** A cell is answered only if the model returns a parseable, finite prediction.
 104 Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign
 105 unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform
 106 guess); we also report the strict upper bound π . Answering is thus part of the score.

107 **Verification record.** For each evaluation cell we retain the trajectory identifier, model and de-
 108 ployment name, prompt, raw response, parser outcome, prediction, prompt and completion token
 109 counts, and score. The released configuration fixes the data-generation solver, held-out seed, horizons,
 110 prompts, timeout, and temperature. Two standards apply here. Local methods are reproducible:
 111 their checkpoints and seeds regenerate the predictions exactly. Hosted models are not, because
 112 a deployment can change or be withdrawn between evaluations, so for them the record is what

remains. We therefore release the per-cell responses rather than summary scores alone, which keeps the reported API evaluation auditable after the deployments it measured are gone.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

Disclosure gap. We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$. Positive values indicate a penalty for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty, we also run a matched experiment (§5.2) that toggles disclosure on identical trajectories.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). Each cell uses one temperature-0 call with a uniform token and timeout budget; no best-of- N aggregation is used. The LLM evaluation (Parts A and B, 3,960 cells) records prompt and completion token counts and has an estimated nominal API cost of \$110.38 under the pricing assumptions in the released analysis script.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU in its standard history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels, resampled to up to 512 points, forecasts to the target horizon (prediction length ≤ 64 steps). The comparison uses `chronos-2` in univariate and multivariate modes; the latter forecasts all $2k$ channels jointly. The paired mode comparison uses 20 trajectories per (k, regime) cell. The earlier `chronos-t5-large` was preliminary; Moirai’s `uni2ts` could not be installed in our environment.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation, which Apple’s MPS backend does not run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU. The primary models are trained on nine trajectories, one for each (k, regime) cell (Appendix B). A nested budget study trains fresh Neural ODE and HNN checkpoints with $\{1, 5, 20\}$ trajectories per eligible cell and seeds $\{101, 102, 103\}$. The Neural ODE uses all nine cells; because damping violates Hamiltonian conservation, the HNN budget study uses only the three undamped normal-regime cells. Every checkpoint is evaluated on the same two held-out seeds.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 generates the data and is included as a numerical reference, not an independent competitor.

Baselines. Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast from the matching (k, regime) training trajectory. The nearest-neighbor baseline uses the same data as the learned models.

On the balanced two-seed subset we also fit SINDy with a degree-two polynomial library and STLSQ, and nonlinear vector autoregression (NVAR) with two delays, degree-two features, and ridge regression. Divergent rollouts are retained as failures and receive the same $\pi/2$ penalty.

Lorenz replication. A secondary Lorenz–63 test compares numerical references, persistence, SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on five

Table 3: Out-of-sample cross-family comparison on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 generates the data and is included as a numerical reference. Methods new in this work are marked $\dagger$; simple baselines are marked $\ddagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimik2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor $\ddagger$	0.725	[0.657, 0.797]	100%	baseline
7	linearized $\ddagger$	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence $\ddagger$	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed $\dagger$	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed $\dagger$	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity $\ddagger$	1.319	[1.241, 1.396]	100%	baseline
18	ltn $\dagger$ (full-scale)	1.356	[1.282, 1.430]	94%	learned

161 five-trajectory held-out seeds. We report training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s;
 162 Appendix C gives the full configuration. HNN and LNN are omitted because Lorenz-63 is dissipative
 163 and non-Hamiltonian.

164 5 Results

165 5.1 Counting failed responses

166 The shared held-out grid contains 360 cells: 180 trajectories at horizons $\{1, 10\}$ s. Every model
 167 is evaluated on that grid. The Answered column in Table 3 gives the fraction of cells with valid
 168 predictions. Confidence intervals are 95% paired bootstrap over trajectories ($B=10^4$); for LLMs they
 169 do not include per-call sampling variation.

170 Table 3 ranks the models on the pendulum under the access conditions in §3.3. The ranking is
 171 conditional on those inputs, this training budget, and this system; Section 5.4 tests whether it transfers.
 172 The known-equation integrators sit at the top as references (RK4 0.059 at a finer step, symplectic
 173 0.205, Euler 0.405), and every data-driven method is well behind them, closer to the $\pi/2$ random
 174 baseline than to the integrators. The lowest-error black-box entry is the frontier LLM kimi (0.538 rad,
 175 99% answered), whose interval excludes the nearest baseline. This establishes its performance for
 176 the recorded June 2026 deployment and evaluation conditions. It does not establish a model-family
 177 ordering, which the Lorenz test rejects.

178 Among the trained models the picture is sharper. After equalizing the hidden-regime inputs, nearest-
 179 neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persis-
 180 tence (1.004) beats all but HNN-rollout (0.972), with the closest intervals overlapping. The stable
 181 learned models score about 0.26 rad on their training trajectories and 0.97–1.14 rad on the held-out
 182 set, so in-sample evaluation understates their error by about a factor of four.

183 Two further data-driven methods appear only on the balanced two-seed subset, so we report them in
 184 Table 4 rather than in the main comparison: sparse identification (SINDy, 0.883 rad on that subset)
 185 falls between the Chronos family and the learned models, and nonlinear vector autoregression (NVAR,
 186 1.248) is among the weakest data-driven methods, near constant velocity.

187 **Effect of missing predictions.** grok answers 82% of the grid and lnn 94%. Their ranks fall when
 188 missing cells receive $\pi/2$. An answered-cells-only mean omits the cases on which these models
 189 failed and gives both a lower reported error.

190 5.2 Matched disclosure test

191 On the shared held-out set the hidden regime has lower error than the disclosed regime (-0.390 rad,
 192 $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524). Read directly, this looks favorable
 193 to withholding constants, but the regimes use different trajectories, and the hidden ones are slower
 194 and less divergent, so the comparison cannot speak to parameter recovery. When we instead toggle
 195 disclosure on identical trajectories (Table 7), the pooled effect reverses: hiding the constants increases
 196 error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with disclosure helping at $k=1$ and at the 1 s horizon and
 197 only a small reverse effect at $k=3$. The unadjusted hidden-regime advantage is thus a difficulty
 198 confound.

199 The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto$
 200 (am, ad) and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from
 201 an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. The
 202 original single-state probe returned mostly Earth-standard defaults, but we treat those outputs as
 203 descriptive and make no system-identification claim from them. The released numerical check verifies
 204 both invariances to below 10^{-13} over one-second trajectories.

205 5.3 Replication across seeds and training budgets

206 We reran the local pendulum subset on an independent held-out seed with five trajectories per
 207 (k, regime) cell; the original seed has 20. Table 4 reports each seed separately and gives their
 208 unweighted mean. The selected ordering is unchanged. The numerical references have the lowest
 209 errors, Chronos-2 precedes the learned models, and HNN-rollout has the lowest error among the
 210 selected learned models. Two seeds do not estimate seed-level uncertainty. The LNN needs a note: its
 211 low coverage was a hardware artifact, not model behavior. Its inference solves a Hessian at each step,
 212 which exhausted the 8 GB GPU on the longest rollouts and returned those cells as failures. Re-running
 213 the LNN on CPU is the same evaluation with the same forecasts; it only recovers the dropped cells,
 214 raising primary-seed coverage from 66% to 94%. We use this CPU evaluation for the LNN on both
 215 seeds (the second answers all 90 cells, equal-seed rate 97%), so its numbers are produced identically
 216 and are comparable across seeds. The cells that still fail are genuine autograd-ODE divergences,
 217 scored at $\pi/2$ like any other failure, and the mean stays high at 1.366 rad.

218 **Training-budget check.** With three training initializations and the two held-out seeds, Neural ODE
 219 error decreases from 1.090 at one trajectory per cell to 0.968 at five and 0.861 at twenty (Table 9).
 220 HNN error on the normal regime falls from 1.269 to 0.974 and then remains 0.981. The HNN and
 221 Neural ODE rows cover different regimes and are not an architecture comparison; within each row,
 222 they show that the nine-trajectory primary result is partly data-limited, while more data alone does
 223 not close the gap to known-equation integration.

224 5.4 Cross-system test: Lorenz-63

225 We test whether the pendulum conclusions transfer to Lorenz-63, a chaotic system with a polynomial
 226 vector field. They do not. Both the identifiability result and the model ordering change.

227 The identifiability verdict flips. On the pendulum the physical constants are not recoverable from an
 228 angle trajectory (§5.2); on Lorenz, sparse identification (SINDy, a degree-two library with STLSQ)
 229 recovers the vector field to normalized MAE below 2×10^{-5} at every horizon and in every seed.
 230 The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is
 231 observed in full state, so its parameters are identifiable, whereas the pendulum’s absolute constants
 232 are not identifiable from angles under any method. Identifiability is a property of the observation
 233 model and the hypothesis class, which is what the check is meant to expose.

234 The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25
 235 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and
 236 0.277 ± 0.063 at horizons 0.1, 0.5, and 1.0 s (mean $\pm$ standard error over seeds), below both Chronos-

237 2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate
238 0.126 ± 0.027 , 0.700 ± 0.102 , 0.732 ± 0.085). NVAR is close at 0.1 s (0.012) but drifts to 0.429 by
239 1.0 s, a persistence baseline trails (0.900 at 1 s), and divergent rollouts are scored at $\pi/2$ as on the
240 pendulum (Table 4). This is the reverse of the pendulum, where the trained models lost to trivial
241 baselines: here a system-trained model beats the zero-shot foundation model. As on the pendulum
242 the families see different inputs (SINDy the library, the Neural ODE target-system training, Chronos
243 a history window), so this is not an architecture race; the cross-system result rejects the pendulum
244 ranking as a general ordering of model families.

245 The reversal does not rest on a favorable seed. The Neural ODE has lower error than both Chronos
246 modes in all 15 seed-by-horizon cells, the separation is roughly four standard errors at each horizon,
247 and SINDy stays below 1.7×10^{-5} in every cell. The identifiability half of the contrast is not a
248 sampling claim at all: the pendulum symmetries are exact and verified to below 10^{-13} , and Lorenz-63
249 lies in SINDy’s library by construction.

250 The Lorenz test includes the local methods for which the comparison is well defined. It excludes the
251 API LLM grid and the Hamiltonian and Lagrangian networks; a Hamiltonian surrogate is misspecified
252 for the dissipative Lorenz flow. Those omissions limit what the test can establish, but not what it can
253 refute: the pendulum result is only interesting if it generalizes, and one system on which the ordering
254 and the identifiability verdict both invert is enough to withdraw that generalization. We claim no
255 more than that, and in particular the Lorenz ordering is not offered as the correct one.

256 5.5 Diagnostics

257 **Behavior by horizon.** Out-of-sample error by horizon appears in Appendix D (Table 12). At 1 s
258 the ordering follows the main comparison: the integrators are near-exact, kimi (0.457) has the lowest
259 black-box error, and the learned models are at 0.86–1.26. At 10 s all models are in the chaotic regime,
260 and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned
261 models reach 1.09–1.44. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is
262 mid-pack here (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder:
263 the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near
264 the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which
265 the families remain separable.

266 **Breakdown by k and regime.** Tables 5 and 6 (Appendix C) report out-of-sample error by number
267 of links and by regime. Error increases monotonically with k for every family (for kimi, $0.11 \rightarrow$
268 $0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family
269 ordering holds within each k . The changed_hidden regime has the lowest error for almost every
270 model (for kimi, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden
271 constants reduces but does not remove the same difficulty effect: hidden-regime errors are now
272 0.71–1.19; the previous values favored models given the true constants. The low-gravity, short-rod
273 constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-
274 regime error contrast is a difficulty confound (§5.2). A cluster bootstrap over complete held-out
275 trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.
276 The original bootstrap’s treatment of correlated cells does not determine the reported gaps.

277 **Training-objective check.** Table 10 reports the effect of the loss with architecture fixed. Under
278 the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower
279 for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the
280 LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the
281 rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under
282 our implementation and protocol the LNN was numerically unstable and generalized poorly out of
283 sample.

284 **Physical-fidelity check.** Hamiltonian and Lagrangian networks target structure preservation, so
285 we evaluate energy drift on the undamped (normal) regime, where true energy is constant. Table 11
286 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded
287 rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned
288 model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic
289 rollout of the HNN leaves drift nearly unchanged. Changing the integrator therefore does not

account for the HNN’s drift. At matched objective, HNN drift is also comparable to Neural ODE drift; the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

6 Discussion

The audit changes five conclusions supported by the raw comparison. Held-out evaluation exposes a roughly fourfold learned-model generalization gap; failure accounting penalizes incomplete coverage; a matched intervention reverses the disclosure effect; exact symmetries rule out absolute parameter recovery; and Lorenz–63 changes the model ordering. These are changes to the scientific claim, not to presentation.

More data improves the Neural ODE from 1.090 to 0.861 rad without closing the gap to known-equation integration; HNN improvement plateaus. On Lorenz, SINDy is nearly exact because the vector field lies in its library, while a system-trained Neural ODE beats zero-shot Chronos-2. Performance therefore depends on the system, observations, hypothesis class, data, and access. A scalar leaderboard does not encode those conditions.

The API LLM evaluation is the only billed run. Its per-cell token records give a nominal cost of \$110.38 for 3,960 forecasts, or about \$0.03 each under the June 2026 deployments and pricing. The local methods incur compute cost without a per-query charge.

7 Limitations

The full LLM grid uses one held-out seed; the local replication adds a second, unequal sample. The budget study uses three training initializations and two held-out seeds. Inputs differ by method as listed in Table 2. The Lorenz test uses five trajectories in each of five held-out seeds and excludes API LLMs, HNN, and LNN; SINDy’s result depends on a library containing the true polynomial degree. Predictions are endpoints, conservation tests cover only the undamped regime, and foundation-model training corpora are not observable.

8 Conclusion

Correct scores can support wrong scientific conclusions. Here, generalization, failure accounting, matched intervention, identifiability, and cross-system transfer each change a claim supported by the raw leaderboard. Scientific forecasting benchmarks should report those checks, their training budgets, and each method’s information access as part of the result. We chose systems whose ground truth is cheap so that each check could be shown to change a conclusion, but the checks are worth most where the reference model is itself imperfect and verification is expensive; carrying them to systems where those constraints bind is what this setup was built to support. We release the prompts, outputs, checkpoints, seeds, and compute records needed to reproduce the audit.

References

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research (TMLR)*; *arXiv:2403.07815*, 2024.
- Johannes G. Borgqvist, Alexander P. Browning, Fredrik Ohlsson, and Ruth E. Baker. Framing global structural identifiability in terms of parameter symmetries. *arXiv preprint arXiv:2410.03757*, 2024.
- Steven L. Brunton, Joshua L. Proctor, and J. Nathan Kutz. Discovering governing equations from data by sparse identification of nonlinear dynamical systems. *Proceedings of the National Academy of Sciences*, 113(15): 3932–3937, 2016. doi: 10.1073/pnas.1517384113.

- 335 Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential
336 equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- 337 Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian
338 neural networks. *ICLR 2020 Workshop on Integration of Deep Neural Models and Differential Equations*;
339 *arXiv:2003.04630*, 2020.
- 340 Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series
341 forecasting. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2310.10688.
- 342 Daniel J. Gauthier, Erik Bollt, Aaron Griffith, and Wendson A. S. Barbosa. Next generation reservoir computing.
343 *Nature Communications*, 12:5564, 2021. doi: 10.1038/s41467-021-25801-2.
- 344 Yonatan Geifman and Ran El-Yaniv. Selective classification for deep neural networks. In *Advances in Neural
345 Information Processing Systems*, volume 30, 2017.
- 346 William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. In *NeurIPS
347 Datasets and Benchmarks Track*, 2021. arXiv:2110.05266.
- 348 Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural
349 Information Processing Systems (NeurIPS)*, 2019.
- 350 Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time
351 series forecasters. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.07820.
- 352 Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh,
353 Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning
354 problems with language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
355 arXiv:2206.14858.
- 356 Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and
357 Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural
358 Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.
- 359 Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training
360 of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*,
361 2024. arXiv:2402.02592.
- 362 Xin Xu, Qiyun Xu, Tong Xiao, Tianhao Chen, Yuchen Yan, Jiaxin Zhang, Shizhe Diao, Can Yang, and Yang
363 Wang. Ugphysics: A comprehensive benchmark for undergraduate physics reasoning with large language
364 models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.00334.
- 365 Yuanzhao Zhang and William Gilpin. Zero-shot forecasting of chaotic systems. In *International Conference on
366 Learning Representations (ICLR)*, 2025. arXiv:2409.15771.

367 A Prompt templates

368 The prompts (from `bench/prompts.py`) are reproduced below.

369 System prompt (direct / no-CoT).

370 You are a careful physicist predicting the future state of a k-pendulum (a chain of
371 k point masses on massless rods, hinged in a planar chain, swinging under gravity
372 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
373 no code fences. Schema:
374 {"theta": [...k floats..], "omega": [...k floats..]}
375 theta_i is in radians (link i measured from straight-down,
376 positive counter-clockwise)
377 and omega_i is angular velocity in rad/s.

378 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,
379 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a
380 single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}.”

381 **User prompt (coordinate modality).**

```
382 Number of links  $k = \{k\}$ 
383 Physical constants:
384    $g$  (gravity,  $m/s^2$ ):  $\{g\}$ 
385    $L$  (rod lengths,  $m$ ):  $\{L\}$ 
386    $m$  (point masses,  $kg$ ):  $\{m\}$ 
387   damping (viscous coefficient on  $\omega$ ):  $\{damping\}$ 
388 Initial state at  $t=0$ :
389    $\theta_0$  (rad):  $\{\theta_0\}$ 
390    $\omega_0$  (rad/s):  $\{\omega_0\}$ 
391 Predict the state at  $t = \{T\}$  seconds.
392 Return  $\theta$  and  $\omega$  as length- $\{k\}$  JSON arrays.
```

393 **Hidden-constants regime.** The constants block above is replaced by:

```
394 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
395 masses, and damping are NOT disclosed. You must infer dynamics from the initial
396 conditions alone.
```

397 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
398 `<answer>...</answer>` tags; the first `{...}` object is JSON-parsed and validated to contain length- k θ and
399 ω arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

400 B Training and compute

401 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a
402 single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
403 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
404 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
405 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
406 trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
407 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
408 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

409 **Reproducibility records.** The repository holds the code, prompts, trained weights, held-out specifications,
410 and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800),
411 the modality and prompting grid, and the Lorenz sweep. The two large sweeps, the training-budget and balanced-
412 seed studies, are published as release assets. Pendulum training uses seed 42; the primary held-out seed is
413 20260620 (180 trajectories), and the selected local subset also uses seed 20260621 (45 trajectories). LLMs
414 are sampled once at temperature 0. The reported hosted-model numbers recompute from the saved records;
415 re-calling the API is a new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

416 C Seed and system replication

417 Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, training seed 42,
418 and held-out seeds 20260620–20260624. Each held-out seed contains five trajectories, giving 25 held-out
419 trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by
420 analytic derivative matching. SINDy uses a degree-two polynomial library and STLSQ; NVAR uses two delays,
421 degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate
422 modes.

423 D Breakdowns by link count and regime

424 Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by
425 number of links (Table 5) and by regime (Table 6); models in leaderboard order. Cells a model never produced
426 (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 3.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k, regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz-63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	seed 20260620	seed 20260621	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

Table 5: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

Table 6: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity/short-rod constants give slower, less-divergent motion. The unadjusted regime contrast in §5.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 7: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

427 E Additional result tables

428 Full per-model values for the horizon breakdown (§5.5) and the predictability horizon (§5.2), together with the
429 regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity
430 studies. The corresponding summary statistics appear in the main text.

Table 8: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

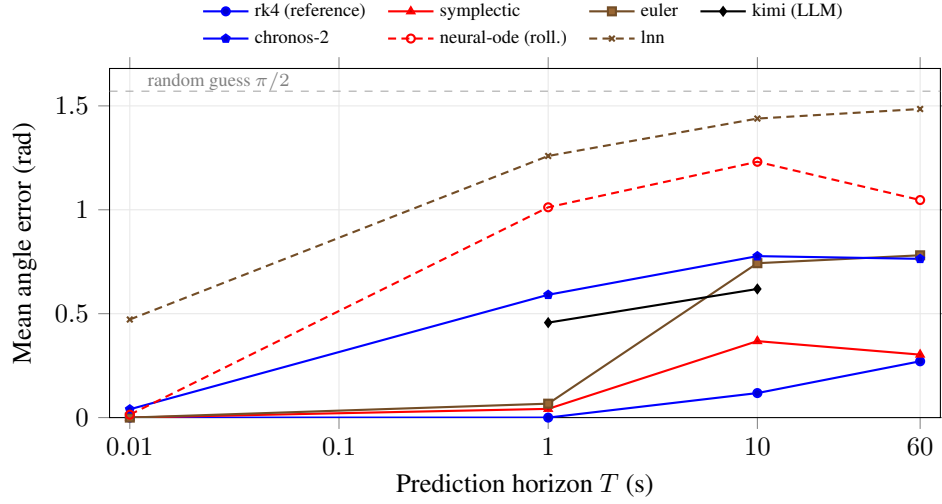


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 9: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 10: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 11: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 12: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.259	1.439

Table 13: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01